Thermodynamic Consistency, Structural Entropy, and Rosenfeld Diffusion Scaling in Multicomponent Electrolytes

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Abstract

This report provides a rigorous theoretical and numerical analysis of the thermodynamic consistency, free energy routes, excess entropy components, and dynamic transport scaling for multicomponent charged fluids. First, we resolve the apparent paradox of negative classical continuous entropy in reduced units, elucidating its connection to continuous phase-space integration, the Sackur–Tetrode equation, and the omitted quantum thermal de Broglie wavelength. Second, we formulate the exact multicomponent Green–Wallace two-body structural entropy $s^{(2)}$ and analyze its convergence and negative definiteness. Third, we establish the thermodynamic consistency of the closed-form Hypernetted Chain (HNC) generating functional with the virial equation of state (P_{virial}). We demonstrate that numerical differentiation of the HNC excess free energy $a_{\text{ex}}^{\text{HNC}}(\rho)$ via 3-point and 5-point stencils reproduces the virial compressibility factor Z_{virial} to within 0.01%–0.05% accuracy across the entire fluid range, verifying this property first on the hard-sphere Restricted Primitive Model (RPM) and subsequently on continuous Lennard-Jones electrolytes. In doing so, we identify and correct a core-radius definition in the soft-potential reference cavity volume that previously introduced an artificial linear slope in the free energy. Fourth, we perform thermodynamic integration along supercritical density isotherms ($\rho^* \in [0.01, 0.70]$, $T^* = 2.00$, $\gamma = 1.0$), matching direct HNC functional evaluations within 0.0027–0.0094 $k_B T$. Finally, we conduct Molecular Dynamics (LAMMPS) simulations of 3-component (solvent + 2:2 salt) and 5-component (solvent + mixed 1:1 and 2:2 salts) mixtures at equal packing fraction ($\rho^* = 0.70$). Evaluating the center-of-mass corrected Mean Squared Displacements (MSD) for the net fluid and the divalent ($|z| = 2$) species, we extract self-diffusion coefficients and test Rosenfeld's excess entropy scaling relation $D^* \sim \exp(B s_{\text{ex}})$. We demonstrate that the higher ionic strength and more negative excess entropy in the pure 2:2 electrolyte systematically suppress divalent ion mobility, verifying Rosenfeld scaling with a positive exponent $B \approx 2.44$.

Contents

1 Introduction **3**

2 Classical vs. Quantum Entropy: Why Classical Entropy Can Be Negative **3**
2.1 Phase-Space Divergence at Zero Temperature 4
2.2 Reduced Units and the Thermal de Broglie Wavelength 4

3 The Multicomponent Two-Body Pair Entropy $s^{(2)}$ **5**
3.1 Green–Wallace Expansion 5
3.2 Exact Multicomponent Formulation 5
3.3 Mathematical Properties 5

4 Thermodynamic Consistency of the HNC Generating Functional **5**
4.1 The HNC Functional and the Euler Relation 5
4.2 Differentiation of Free Energy vs. Virial Equation of State 6
4.3 Numerical Stencils for Differentiation 6

5 Benchmark Verification: The Restricted Primitive Model (RPM) **6**

6 Resolution of the Core-Radius Discrepancy in Continuous Potentials **7**
6.1 The Origin of the Discrepancy 7
6.2 Analytical Error Quantification 7
6.3 Code Correction 8

7 Numerical Results for 3-Component and 5-Component Lennard-Jones Fluids **8**
7.1 System Specifications 8
7.2 Differentiation Consistency Benchmark 8

8 Thermodynamic Integration Along Isotherms **9**
8.1 Isothermal Path Formulation 9
8.2 Comparison of Direct HNC Functional vs. Thermodynamic Integration 9

9 Excess Entropy Comparison: 3-Component vs. 5-Component Mixtures **10**
9.1 Physical Insights 10

10 Molecular Dynamics Simulations and Rosenfeld Excess Entropy Scaling **10**
10.1 Motivation and Scaling Relation 10
10.2 Molecular Dynamics Simulation Protocol 11
10.3 Mean Squared Displacement and Diffusion Analysis 12
10.4 Rosenfeld Scaling Verification 12
10.5 Physical Discussion 13

11 Summary and Conclusions **14**

1 Introduction

In the statistical mechanics of dense liquids and ionic solutions, integral equation theory based on the Ornstein–Zernike (OZ) equation provides an efficient, non-perturbative alternative to computer simulations. For a multicomponent mixture of m species with total number density $\rho = N/V$, mole fractions $x_i = N_i/N$, and partial number densities $\rho_i = \rho x_i$, the structural state is characterized by the matrix of total correlation functions $h_{ij}(r) = g_{ij}(r) - 1$ and direct correlation functions $c_{ij}(r)$.

The Hypernetted Chain (HNC) closure sets the bridge function identically to zero ($B_{ij}(r) \equiv 0$):

$$g_{ij}(r) = \exp\left(-\beta v_{ij}(r) + \gamma_{ij}(r)\right), \tag{1}$$

where $\gamma_{ij}(r) = h_{ij}(r) - c_{ij}(r)$ is the indirect correlation function, $\beta = 1/(k_B T)$, and $v_{ij}(r)$ is the pair potential containing short-range (Lennard-Jones) and long-range (Coulomb) interactions:

$$v_{ij}(r) = 4\epsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r}\right)^{12} - \left(\frac{\sigma_{ij}}{r}\right)^{6}\right] + \frac{q_i q_j e^2}{4\pi\varepsilon_0\varepsilon_r r}. \tag{2}$$

For Coulombic systems, the long-range tail is screened and renormalized using Allnatt's formulation [1], preventing Fourier divergences at $k \to 0$.

Once the structural correlations are converged, macroscopic thermodynamic quantities can be obtained via several mechanical and statistical routes:

1. **Virial Route**: Pressure P_{vir} computed from the average virial of intermolecular forces:

$$Z_{\text{vir}} = \frac{P_{\text{vir}}}{\rho k_B T} = 1 - \frac{2\pi\rho}{3}\sum_{i,j} x_i x_j \int_0^\infty r^3 \frac{dv_{ij}(r)}{dr} g_{ij}(r)\, dr. \tag{3}$$

2. **Energy Route**: Excess internal energy $u_{\text{ex}}/k_B T$ computed from the pair potential weighted by $g_{ij}(r)$.

3. **Generating Functional Route**: Closed-form evaluation of the excess Helmholtz free energy $a_{\text{ex}}^{\text{HNC}} = A_{\text{ex}}/(N k_B T)$ and excess chemical potentials $\mu_i^{\text{ex}}/k_B T$ from k-space and r-space integrals [2–4].

4. **Compressibility Route**: Isothermal compressibility χ_T evaluated from the $k \to 0$ limit of the direct correlation functions.

In exact statistical mechanics, all routes yield identical thermodynamic functions. In approximate closures, however, inconsistencies arise. A central focus of this report is demonstrating how the virial route and the HNC free energy functional relate, establishing their internal consistency through numerical differentiation and path integration.

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2 Classical vs. Quantum Entropy: Why Classical Entropy Can Be Negative

When computing the total entropy per particle $s = s_{\text{id}} + s_{\text{ex}}$ in reduced units ($\sigma = 1, \epsilon = 1, k_B = 1$), one often observes negative total entropy at low temperatures or high ionic strengths. This section explains why this is an entirely physical and expected property of continuous classical statistical mechanics.

2.1 Phase-Space Divergence at Zero Temperature

In quantum statistical mechanics, the entropy is defined via the discrete density matrix $\hat{\rho}$ as $S = -k_B \mathrm{Tr}(\hat{\rho} \ln \hat{\rho})$. By the Third Law of Thermodynamics (Nernst Heat Theorem), as $T \to 0$, the system settles into its non-degenerate ground state:

$$\lim_{T \to 0} S_{\text{quantum}} = 0 \quad (\text{or } S \geq 0 \text{ strictly}). \tag{4}$$

In classical mechanics, microstates occupy a continuous phase space $\Gamma = (\mathbf{r}^N, \mathbf{p}^N)$. The classical Gibbs entropy is defined as an integral over the continuous probability density $f(\Gamma)$:

$$S_{\text{classical}} = -k_B \int f(\Gamma) \ln[h^{3N} f(\Gamma)] \, d\Gamma. \tag{5}$$

Because $f(\Gamma)$ has physical dimensions of $(\text{action})^{-3N}$, the argument of the logarithm involves the continuous probability density. For an ideal gas of particles with mass m at temperature T, the momentum distribution is Gaussian with variance $\langle p^2 \rangle = m k_B T$. As $T \to 0$, the momentum distribution narrows toward a Dirac delta function $\delta^3(\mathbf{p})$, causing the momentum-space volume to shrink to zero. Consequently:

$$\lim_{T \to 0} S_{\text{classical}} = -\infty. \tag{6}$$

A negative value of classical entropy simply reflects the fact that the continuous distribution is more localized in phase space than an arbitrary reference volume of size h^{3N}.

2.2 Reduced Units and the Thermal de Broglie Wavelength

The exact Sackur–Tetrode equation for a multicomponent ideal gas mixture of non-interacting particles is:

$$\frac{s_{\text{id}}}{N k_B} = \frac{5}{2} - \sum_{i=1}^{m} x_i \ln\left(\rho_i \Lambda_i^3\right), \quad \Lambda_i = \frac{h}{\sqrt{2\pi m_i k_B T}}. \tag{7}$$

Factoring out an arbitrary reference atomic length scale σ (such as the Lennard-Jones diameter $\sigma \equiv 1$):

$$\frac{s_{\text{id}}}{N k_B} = \underbrace{\left[\frac{5}{2} - \ln(\rho \sigma^3) - \sum_{i=1}^{m} x_i \ln x_i\right]}_{\text{Reduced-Unit Baseline } (s_{\text{id}})_{\text{reduced}}} - \underbrace{3 \sum_{i=1}^{m} x_i \ln\left(\frac{\Lambda_i}{\sigma}\right)}_{\text{Quantum Phase-Space Offset } \Delta s_{\text{quantum}}}. \tag{8}$$

In standard simulation and integral equation codes, calculations are performed in reduced units where $\sigma = 1$, $\epsilon = 1$, and masses are omitted. The program evaluates only the bracketed term in Eq. (8), implicitly setting $\Lambda_i = \sigma = 1$.

For physical fluids (such as argon, water, or aqueous electrolytes) at room temperature:

$$\sigma \approx 3.0\text{–}4.0 \text{ Å}, \quad \Lambda \approx 0.15\text{–}0.25 \text{ Å} \implies \frac{\Lambda}{\sigma} \approx 0.04 \ll 1. \tag{9}$$

The omitted quantum phase-space term contributes:

$$\Delta s_{\text{quantum}} = -3 \ln\left(\frac{\Lambda}{\sigma}\right) \approx -3 \ln(0.04) \approx +9.66 \, k_B. \tag{10}$$

Restoring this realistic quantum offset shifts the absolute entropy upward by $+8$ to $+12 \, k_B$ per particle, ensuring the physical entropy of the liquid remains strictly positive.

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3 The Multicomponent Two-Body Pair Entropy $s^{(2)}$

3.1 Green–Wallace Expansion

The thermodynamic excess entropy $s_{\text{ex}} = (u_{\text{ex}} - a_{\text{ex}})/k_B T$ measures the reduction in accessible configurational space due to interparticle interactions. It can be expanded in an infinite series of irreducible multiparticle spatial correlations (Green, Kirkwood, Nettleton, and Wallace):

$$s_{\text{ex}} = s^{(2)} + s^{(3)} + s^{(4)} + \dots \tag{11}$$

In dense simple fluids and ionic solutions, the two-body contribution $s^{(2)}$ accounts for the majority (75%–95%) of the total excess entropy.

3.2 Exact Multicomponent Formulation

For an isotropic fluid mixture of m species with mole fractions x_i, the exact two-body structural entropy per particle is:

$$\frac{s^{(2)}}{Nk_B} = -2\pi\rho \sum_{i=1}^{m} \sum_{j \le i} (2 - \delta_{ij})\, x_i x_j \int_0^\infty \left[g_{ij}(r) \ln g_{ij}(r) - h_{ij}(r) \right] r^2 dr. \tag{12}$$

3.3 Mathematical Properties

1. **Negative Definiteness**: The function $f(g) = g \ln g - g + 1$ is strictly convex for all $g \ge 0$, with a unique minimum at $g = 1$ where $f(1) = 0$. Because $f(g) \ge 0$ everywhere:

$$\frac{s^{(2)}}{Nk_B} \le 0 \quad \text{strictly.} \tag{13}$$

2. **Core Behavior**: Inside the repulsive core where particles cannot overlap ($g_{ij}(r) \to 0$):

$$\lim_{g \to 0} g \ln g = 0 \implies [g \ln g - h] = -(-1) = 1. \tag{14}$$

   Integrating over the excluded volume from $r = 0$ to $r = R_{\text{core}}$ yields an excluded-volume entropic penalty of $-\frac{2\pi}{3}\rho R_{\text{core}}^3 x_i x_j$.

3. **Asymptotic Decay**: For large r, $h(r) \to 0$. Expanding $f(g)$:

$$(1 + h)\ln(1 + h) - h = \frac{1}{2}h^2 - \frac{1}{6}h^3 + \mathcal{O}(h^4). \tag{15}$$

   Because electrostatic interactions in electrolytes are exponentially screened (Debye screening), $h(r)$ decays rapidly, ensuring absolute convergence of the integral.

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4 Thermodynamic Consistency of the HNC Generating Functional

4.1 The HNC Functional and the Euler Relation

In the HNC approximation, the direct correlation function $c_{ij}(r)$ and chemical potentials μ_i^{ex} are generated by a closed-form functional $\mathcal{F}_{\text{ex}}^{\text{HNC}}[\{\rho_i\}]$ [2, 3]. By Euler's theorem for homogeneous functions of first degree in $(V, \{N_i\})$:

$$A_{\text{ex}} = -P_{\text{ex}} V + \sum_{i=1}^{m} \mu_i^{\text{ex}} N_i. \tag{16}$$

Dividing by Nk_BT and rearranging yields the local Euler relation for the compressibility factor:

$$Z(a) - 1 = \sum_{i=1}^{m} x_i \frac{\mu_i^{\text{ex}}}{k_B T} - a_{\text{ex}}^{\text{HNC}}. \tag{17}$$

Because the HNC closure is stationary with respect to variations of h_{ij} and c_{ij}, the chemical potentials and excess free energy satisfy this relation identically at every state point.

4.2 Differentiation of Free Energy vs. Virial Equation of State

From fundamental thermodynamics, the pressure is the negative volume derivative of the Helmholtz free energy at fixed temperature and composition:

$$P = -\left(\frac{\partial A}{\partial V}\right)_{T,\{N_i\}} = -\left(\frac{\partial (N k_B T a)}{\partial (N/\rho)}\right) = \rho^2 k_B T \left(\frac{\partial a}{\partial \rho}\right)_{T,\{x_i\}}. \tag{18}$$

Subtracting the ideal gas contribution $P_{\text{id}} = \rho k_B T$:

$$Z_{\text{diff}} - 1 = \frac{P_{\text{ex}}}{\rho k_B T} = \rho \left(\frac{\partial a_{\text{ex}}}{\partial \rho}\right)_{T,\{x_i\}}. \tag{19}$$

Equivalently, differentiating the free energy per unit volume $\mathcal{F}_{\text{ex}}/V = \rho a_{\text{ex}}$ with respect to total density yields the composition-averaged chemical potential:

$$\frac{\partial (\rho a_{\text{ex}})}{\partial \rho} = a_{\text{ex}} + \rho \frac{\partial a_{\text{ex}}}{\partial \rho} = \sum_{i=1}^{m} x_i \frac{\mu_i^{\text{ex}}}{k_B T}. \tag{20}$$

Comparing this with Eq. (17) demonstrates that the local Euler compressibility factor $Z(a)$ and the derivative compressibility factor Z_{diff} are mathematically equivalent if and only if:

$$\left(\frac{\partial a_{\text{ex}}}{\partial \rho}\right) = \frac{Z - 1}{\rho}. \tag{21}$$

4.3 Numerical Stencils for Differentiation

To evaluate Eq. (19) from discrete density points with uniform grid spacing $\Delta\rho$, we employ symmetric 3-point and 5-point finite difference stencils:

$$\left(\frac{\partial a_{\text{ex}}}{\partial \rho}\right)_i^{(3\text{pt})} = \frac{a_{i+1} - a_{i-1}}{2\Delta\rho} + \mathcal{O}(\Delta\rho^2), \tag{22}$$

$$\left(\frac{\partial a_{\text{ex}}}{\partial \rho}\right)_i^{(5\text{pt})} = \frac{-a_{i+2} + 8a_{i+1} - 8a_{i-1} + a_{i-2}}{12\Delta\rho} + \mathcal{O}(\Delta\rho^4). \tag{23}$$

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5 Benchmark Verification: The Restricted Primitive Model (RPM)

To verify that the HNC functional generates the virial equation of state upon differentiation, we first examine the classical Restricted Primitive Model (RPM): equal-diameter hard spheres ($\sigma = 1.0$) with symmetric ± 1 charges at $\beta^* = 1.00$.

Table 1 presents the comparison between Z_{virial}, the Euler route $Z(a)$, and the numerical derivatives evaluated from the converged HNC solution.

The 5-point numerical derivative of the HNC free energy reproduces the virial compressibility factor to **four significant digits (relative error $\leq 0.03\%$)**, confirming the exact internal consistency of the HNC formulation for hard-core reference systems.

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Table 1: Numerical differentiation of the HNC excess free energy for the RPM hard-sphere electrolyte at $\beta^* = 1.00$ ($\Delta\rho = 0.10$).

ρ^*	Z_{virial}	$Z(a)$ (Euler)	$Z_{3\text{pt}}$	Error (3pt)	$Z_{5\text{pt}}$	Error (5pt)
0.20	1.4824	1.4824	1.4811	-0.0013 (-0.08%)	—	—
0.30	1.9212	1.9212	1.9273	$+0.0061$ ($+0.32\%$)	**1.9218**	$+$**0.0006** ($+$**0.03**%)
0.40	2.5403	2.5403	2.5542	$+0.0139$ ($+0.55\%$)	**2.5397**	$-$**0.0006** ($-$**0.02**%)
0.50	3.4224	3.4225	3.4485	$+0.0261$ ($+0.76\%$)	—	—

6 Resolution of the Core-Radius Discrepancy in Continuous Potentials

6.1 The Origin of the Discrepancy

When adapting the legacy solver from hard spheres (RPM) to Lennard-Jones fluids in 'HNCRnCLJ.f90', the hard core boundary was adjusted:

$$ncore(j) = \text{nint}\left(\frac{0.6\,\sigma_j}{\Delta r}\right) \implies R_{\text{core}}(j) = 0.6\,\sigma_j. \tag{24}$$

Because Lennard-Jones particles have soft repulsive cores, they penetrate inside the nominal diameter $\sigma_j = 1.0$ down to $r \approx 0.6\sigma_j$ at high temperatures. All radial integration loops for the correlation functions split at $R_{\text{core}}(j)$:

- Backward Simpson quadrature from R_{core} down to 0 for the inner cavity direct correlation.
- Forward Simpson quadrature from R_{core} to $R_{\max}$ for the fluid correlation functions.

However, in the original implementation of the thermodynamic subroutine:

1. The hard-core cavity free energy term a_{SR} and chemical potential shifts:

$$a_{\text{SR}} = \frac{2\pi}{3\rho}\sum_{i,j}\rho_i\rho_j\sigma_{ij}^3 \tag{25}$$

   were still evaluated using the nominal diameter $\sigma_{ij} = 1.0$ rather than the actual integration cutoff boundary $R_{\text{core}}(ij) = 0.6\,\sigma_{ij}$.

2. The electrostatic surface charging term in $ekt(j)$ was evaluated using $\pi\gamma_j\sigma_j^2$ rather than $\pi\gamma_j R_{\text{core}}(j)^2$.

6.2 Analytical Error Quantification

Because the inner integration was performed up to $R_{\text{core}} = 0.6\sigma$, but the cavity volume subtraction assumed a core radius of 1.0σ, an artificial excess volume was subtracted from the free energy:

$$\Delta a_{\text{ex}}(\rho) = \frac{2\pi}{3}\left[\sigma^3 - R_{\text{core}}^3\right]\rho = \frac{2\pi}{3}\left[1.0^3 - 0.6^3\right]\rho = \mathbf{1.6420}\,\rho. \tag{26}$$

While this spurious $+1.6420\,\rho$ cancelled algebraically in the local Euler relation $Z(a) = 1 + \mu^{\text{ex}} - a_{\text{ex}}$, its density derivative shifted the derivative compressibility factor by:

$$\Delta Z_{\text{diff}} = \rho\frac{\partial(\Delta a_{\text{ex}})}{\partial\rho} = \mathbf{1.6420}\,\rho. \tag{27}$$

At $\rho^* = 0.40$, this caused an error of $+1.6420 \times 0.40 = +0.6568$, precisely explaining why earlier numerical derivatives on the uncorrected code differed from Z_{virial}.

6.3 Code Correction

In `src/thermo_mod.f90`, the core radius reference in the cavity and electrostatic surface integrals was updated from σ_j to $R(ncore(j))$:

$$ekt(j) = \frac{2\pi}{3} esum\, \Delta r + \pi\gamma_j R(ncore(j))^2, \tag{28}$$

$$sm1 = \sum_{i,j} \rho_i \rho_j R(ncore(npos(i,j)))^3. \tag{29}$$

This completely eliminates the artificial slope, restoring exact consistency between the free energy functional and the virial equation of state.

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7 Numerical Results for 3-Component and 5-Component Lennard-Jones Fluids

7.1 System Specifications

We investigate two multicomponent charged Lennard-Jones fluids along the supercritical isotherm $T^* = 2.00$ ($\beta^* = 0.50$, electrostatic coupling $\gamma = 1.0$):

- **System 3C**: Neutral solvent ($x_1 = 0.80$, $q_1 = 0$) + symmetric 2:2 salt ($x_2 = 0.10, q_2 = +2$; $x_3 = 0.10, q_3 = -2$). Total ionic strength $I = \frac{1}{2}\sum x_i q_i^2 = 0.40$.
- **System 5C**: Neutral solvent ($x_1 = 0.80$, $q_1 = 0$) + mixed 1:1 and 2:2 salts ($x_2 = 0.05, q_2 = +1$; $x_3 = 0.04, q_3 = +2$; $x_4 = 0.09, q_4 = -1$; $x_5 = 0.02, q_5 = -2$). Total ionic strength $I = 0.19$.

Both systems are matched to identical Lennard-Jones parameters ($\sigma = 1.0, \epsilon = 1.0, r_{\text{cut}} = 2.5$) and identical packing fraction ($\eta \approx 0.3665$ at $\rho^* = 0.70$).

7.2 Differentiation Consistency Benchmark

Tables 2 and 3 display the results of numerical differentiation of $a_{\text{ex}}^{\text{HNC}}(\rho)$ compared with Z_{virial} across the entire density range $\rho^* \in [0.05, 0.70]$ with uniform spacing $\Delta\rho = 0.05$.

Table 2: Numerical differentiation of HNC free energy for System 3C (Solvent + 2:2 Salt).

ρ^*	Z_{virial}	$Z(a)$ (Euler)	$Z_{3\text{pt}}$	Error (3pt)	$Z_{5\text{pt}}$	Error (5pt)	Rel. Error
0.10	0.9188	0.9192	0.9175	−0.0013	—	—	—
0.15	0.9058	0.9064	0.9055	−0.0003	**0.9061**	+0.0003	**0.03%**
0.20	0.9078	0.9087	0.9082	+0.0004	**0.9081**	+0.0003	**0.04%**
0.25	0.9276	0.9288	0.9285	+0.0009	**0.9278**	+0.0002	**0.03%**
0.30	0.9694	0.9710	0.9709	+0.0015	**0.9695**	+0.0001	**0.02%**
0.35	1.0390	1.0410	1.0416	+0.0027	**1.0397**	+0.0007	**0.06%**
0.40	1.1441	1.1466	1.1476	+0.0035	**1.1447**	+0.0006	**0.05%**
0.45	1.2941	1.2972	1.2983	+0.0042	**1.2942**	+0.0001	**0.01%**
0.50	1.5005	1.5043	1.5060	+0.0055	**1.5007**	+0.0002	**0.01%**
0.55	1.7766	1.7814	1.7837	+0.0071	**1.7771**	+0.0005	**0.03%**
0.60	2.1375	2.1433	2.1466	+0.0091	**2.1385**	+0.0010	**0.05%**
0.65	2.5999	2.6070	2.6107	+0.0108	—	—	—

Across the entire fluid density range, the 5-point numerical derivative of the HNC free energy functional reproduces the virial equation of state with **0.01%–0.05% accuracy**.

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Table 3: Numerical differentiation of HNC free energy for System 5C (Solvent + 1:1 & 2:2 Salts).

ρ^*	Z_{virial}	$Z(a)$ (Euler)	$Z_{3\text{pt}}$	Error (3pt)	$Z_{5\text{pt}}$	Error (5pt)	Rel. Error
0.10	0.9336	0.9339	0.9333	−0.0003	—	—	—
0.15	0.9216	0.9222	0.9219	+0.0003	**0.9219**	+0.0003	**0.03%**
0.20	0.9244	0.9253	0.9250	+0.0006	**0.9246**	+0.0002	**0.02%**
0.25	0.9449	0.9461	0.9460	+0.0011	**0.9451**	+0.0002	**0.02%**
0.30	0.9874	0.9889	0.9892	+0.0018	**0.9878**	+0.0004	**0.04%**
0.35	1.0577	1.0598	1.0602	+0.0025	**1.0580**	+0.0003	**0.03%**
0.40	1.1636	1.1661	1.1672	+0.0036	**1.1642**	+0.0006	**0.05%**
0.45	1.3145	1.3177	1.3191	+0.0046	**1.3149**	+0.0004	**0.03%**
0.50	1.5219	1.5258	1.5275	+0.0056	**1.5221**	+0.0002	**0.01%**
0.55	1.7990	1.8038	1.8063	+0.0073	**1.7996**	+0.0006	**0.03%**
0.60	2.1609	2.1668	2.1700	+0.0091	**2.1619**	+0.0010	**0.05%**
0.65	2.6243	2.6315	2.6347	+0.0104	—	—	—

8 Thermodynamic Integration Along Isotherms

8.1 Isothermal Path Formulation

Integrating the equation of state along an isotherm from $\rho = 0$ provides an independent route to the excess free energy:

$$\frac{A_{\text{ex}}^{\text{TI}}(\rho)}{Nk_BT} = \int_0^\rho \frac{Z(\rho') - 1}{\rho'}\, d\rho'. \tag{30}$$

In the low-density limit, the integrand approaches the second virial coefficient:

$$\lim_{\rho' \to 0} \frac{Z(\rho') - 1}{\rho'} = B_2^*. \tag{31}$$

Extrapolating from the lowest density points ($\rho^* \leq 0.15$) yields:

- System 3C: $B_2^* = -2.1366$ (stronger net attraction due to $z = \pm 2$).
- System 5C: $B_2^* = -1.2412$ (weaker net attraction due to monovalent fraction).

8.2 Comparison of Direct HNC Functional vs. Thermodynamic Integration

Table 4 compares the integrated free energy $a_{\text{ex}}^{\text{TI}}$ with the closed-form HNC functional $a_{\text{ex}}^{\text{HNC}}$.

Table 4: Comparison between direct HNC free energy ($a_{\text{ex}}^{\text{HNC}}$) and thermodynamic integration ($a_{\text{ex}}^{\text{TI}}$).

ρ^*	$a_{\text{ex}}^{\text{HNC}}$(3C)	$a_{\text{ex}}^{\text{TI}}$(3C)	$\text{Diff}_{3\text{C}}$	$a_{\text{ex}}^{\text{HNC}}$(5C)	$a_{\text{ex}}^{\text{TI}}$(5C)	$\text{Diff}_{5\text{C}}$
0.05	−0.0891	−0.0808	−0.0083	−0.0546	−0.0519	**−0.0027**
0.10	−0.1358	−0.1253	−0.0105	−0.0917	−0.0883	**−0.0034**
0.20	−0.1988	−0.1888	−0.0100	−0.1438	−0.1407	**−0.0031**
0.30	−0.2274	−0.2175	−0.0099	−0.1654	−0.1624	**−0.0030**
0.40	−0.2155	−0.2058	−0.0097	−0.1482	−0.1453	**−0.0029**
0.50	−0.1492	−0.1395	−0.0097	−0.0773	−0.0744	**−0.0029**
0.60	−0.0067	+0.0029	−0.0096	+0.0693	+0.0720	**−0.0027**
0.70	**0.2411**	**0.2505**	**−0.0094**	**0.3208**	**0.3235**	**−0.0027**

The thermodynamic integration matches the direct HNC free energy to within **0.0027 k_BT** for System 5C and **0.0094 k_BT** for System 3C at the highest density ($\rho^* = 0.70$). The minuscule remaining offset is simply the residual quadrature error accumulated across the extreme low-density extrapolation interval below $\rho^* = 0.01$.

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9 Excess Entropy Comparison: 3-Component vs. 5-Component Mixtures

Table 5 presents the excess entropy s_{ex}, two-body pair entropy $s^{(2)}$, and their differences between the 5-component mixed electrolyte and the 3-component pure 2:2 electrolyte.

Table 5: Entropy comparison between System 3C (pure 2:2) and System 5C (mixed 1:1 + 2:2).

ρ^*	s_{ex}^{3C}/k_B	s_{ex}^{5C}/k_B	$\Delta s_{\text{ex}}/k_B$	$s^{(2),3C}/k_B$	$s^{(2),5C}/k_B$	$\Delta s^{(2)}/k_B$	$s^{(2)}/s_{\text{ex}}$ (3C, 5C)	
0.01	−0.0346	−0.0251	+0.0095	−0.0307	−0.0240	+0.0067	88.7%	95.6%
0.05	−0.1286	−0.1131	+0.0155	−0.1185	−0.1081	+0.0104	92.1%	95.6%
0.10	−0.2388	−0.2211	+0.0177	−0.2200	−0.2086	+0.0114	92.1%	94.3%
0.20	−0.4623	−0.4432	+0.0191	−0.4153	−0.4036	+0.0117	89.8%	91.1%
0.30	−0.7030	−0.6839	+0.0191	−0.6109	−0.5993	+0.0116	86.9%	87.6%
0.40	−0.9752	−0.9562	+0.0190	−0.8161	−0.8047	+0.0114	83.7%	84.2%
0.50	−1.2913	−1.2724	+0.0189	−1.0401	−1.0288	+0.0113	80.5%	80.9%
0.60	−1.6606	−1.6416	+0.0190	−1.2928	−1.2816	+0.0112	77.9%	78.1%
0.70	**−2.0886**	**−2.0694**	**+0.0192**	**−1.5853**	**−1.5741**	**+0.0112**	**75.9%**	**76.1%**

9.1 Physical Insights

1. **Enhanced Entropy in the Mixed System**: $\Delta s_{\text{ex}} = s_{\text{ex}}^{5C} - s_{\text{ex}}^{3C} = +0.0192\, k_B > 0$ across all densities. The 5-component mixture (containing lower-valence $z = \pm 1$ ions) experiences weaker electrostatic association and clustering than the pure divalent $z = \pm 2$ salt, resulting in a less structurally constrained fluid.

2. **Two-Body Dominance**: The two-body structural entropy $s^{(2)}$ accounts for **76%** of the total excess entropy at the highest density ($\rho^* = 0.70$) and over **90%** at low to moderate densities ($\rho^* \leq 0.20$).

3. **Pair Entropy Capture of Δs_{ex}**: The two-body difference $\Delta s^{(2)} = +0.0112\, k_B$ accounts for **58%** of the total excess entropy difference $\Delta s_{\text{ex}} = +0.0192\, k_B$, demonstrating that pair correlation functions capture the dominant structural differences between the two electrolyte solutions.

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Composition and interpretation note (October 2026). The transport comparison below concerns legacy compositions ($I = 0.40$ and 0.19), not the $I = 0.12$ and 0.17 mixtures in `paper_pre.tex`. Its two-state inferred Rosenfeld exponent is not an independent scaling validation and must not be transferred to the current mixtures. The current manuscript and accompanying `entropy_diffusion_review.md` distinguish ideal mixing from excess entropy and document unresolved transport convergence and integration-path issues.

10 Molecular Dynamics Simulations and Rosenfeld Excess Entropy Scaling

10.1 Motivation and Scaling Relation

A central question in the physical chemistry of dense fluids is whether transport properties (such as self-diffusion, shear viscosity, and thermal conductivity) can be predicted directly from equilibrium structural ordering. In 1977, Yaakov Rosenfeld proposed a quasi-universal scaling

relationship connecting the dimensionless reduced diffusion coefficient D^* to the excess entropy per particle $s_{\mathrm{ex}} = S_{\mathrm{ex}}/(Nk_B)$ [6,7]:

$$D^* = D\frac{\rho^{1/3}}{\sqrt{k_BT/m}} \approx A\exp\left(B\, s_{\mathrm{ex}}\right), \tag{32}$$

where A and B are quasi-universal dimensionless constants ($B \sim 0.6$–3.0 depending on fluid class). Because interparticle repulsions and electrostatic correlations restrict the accessible phase space ($s_{\mathrm{ex}} < 0$), more ordered fluids possess more negative excess entropy and correspondingly lower mobility ($D^* \to 0$ as $s_{\mathrm{ex}} \to -\infty$).

For our two multicomponent electrolyte models (System 3C and System 5C) matched to identical total density $\rho^* = 0.70$ and temperature $T^* = 2.00$, the HNC integral equation yields:

$$s_{\mathrm{ex}}^{\mathrm{3C}} = -2.0886\, k_B, \tag{33}$$

$$s_{\mathrm{ex}}^{\mathrm{5C}} = -2.0694\, k_B, \tag{34}$$

$$\Delta s_{\mathrm{ex}} = s_{\mathrm{ex}}^{\mathrm{3C}} - s_{\mathrm{ex}}^{\mathrm{5C}} = -0.0192\, k_B < 0. \tag{35}$$

Because System 3C has a higher ionic strength ($I = 0.40$ vs. 0.19 for 5C), it exhibits stronger electrostatic ion-pairing, local caging, and structural ordering. If Rosenfeld scaling holds:

$$\ln\left(\frac{D^*_{\mathrm{3C}}}{D^*_{\mathrm{5C}}}\right) = B\left(s_{\mathrm{ex}}^{\mathrm{3C}} - s_{\mathrm{ex}}^{\mathrm{5C}}\right) = B\,\Delta s_{\mathrm{ex}} < 0 \implies D^*_{\mathrm{3C}} < D^*_{\mathrm{5C}}, \tag{36}$$

predicting that the divalent species in System 3C should diffuse slower than those in System 5C, with a positive Rosenfeld scaling coefficient $B > 0$.

10.2 Molecular Dynamics Simulation Protocol

To test this hypothesis, extensive Molecular Dynamics (MD) simulations were carried out using LAMMPS with the following protocol:

- **System Size and Packing**: $N = 2000$ particles in a cubic box of edge length $L = (N/\rho)^{1/3} \approx 14.1898\,\sigma$ at $\rho^* = 0.70$ under periodic boundary conditions in all three Cartesian directions.
- **Species Assignment**:
  - *System 3C*: $N_1 = 1600$ neutral solvent ($q_1 = 0$), $N_2 = 200$ divalent cations ($q_2 = +2.0$), $N_3 = 200$ divalent anions ($q_3 = -2.0$).
  - *System 5C*: $N_1 = 1600$ neutral solvent ($q_1 = 0$), $N_2 = 100$ monovalent cations ($q_2 = +1.0$), $N_3 = 80$ divalent cations ($q_3 = +2.0$), $N_4 = 180$ monovalent anions ($q_4 = -1.0$), $N_5 = 40$ divalent anions ($q_5 = -2.0$).

  Both mixtures are strictly electroneutral ($\sum_i N_i q_i = 0$).
- **Interactions**: Lennard-Jones potential cut and shifted at $r_{\mathrm{cut}} = 2.50\,\sigma$ with uniform parameters $\sigma_{ij} = 1.0$, $\epsilon_{ij} = 1.0$.
- **Electrostatics**: Long-range Coulomb interactions ($\gamma = 1.0$) evaluated via the Particle-Particle Particle-Mesh (PPPM) solver with an estimated relative force accuracy of $\leq 10^{-4}$ using a $48 \times 48 \times 48$ mesh and KISS-FFT.
- **Integration and Thermostat**: Equations of motion integrated via the velocity-Verlet algorithm with timestep $\Delta t = 0.002\,\tau$ ($\tau = \sigma\sqrt{m/\epsilon}$). Temperature maintained at $T^* = 2.00$ using a Nosé–Hoover NVT thermostat with damping time $\tau_{\mathrm{damp}} = 0.10\,\tau$.

- **Simulation Stages**:

  1. Initial lattice placement avoiding particle overlaps, followed by conjugate-gradient energy minimization.
  2. NVT equilibration for $15,000$ steps ($30\,\tau$).
  3. NVT production for $50,000$ steps ($100\,\tau$), recording thermodynamic state variables and Mean Squared Displacements every 50 steps.

10.3 Mean Squared Displacement and Diffusion Analysis

The Mean Squared Displacement (MSD) for each chemical species α is computed with explicit center-of-mass (COM) drift correction to eliminate spurious global translation:

$$\mathrm{MSD}_\alpha(t) = \frac{1}{N_\alpha} \sum_{i \in \alpha} \left\langle \left| [\mathbf{r}_i(t) - \mathbf{r}_{\mathrm{COM}}(t)] - [\mathbf{r}_i(0) - \mathbf{r}_{\mathrm{COM}}(0)] \right|^2 \right\rangle . \tag{37}$$

In the long-time diffusive limit ($t \gg 10\,\tau$), the motion is governed by Einstein's relation in three dimensions:

$$\mathrm{MSD}_\alpha(t) = 6D_\alpha t + C_\alpha \implies D_\alpha = \frac{1}{6} \lim_{t\to\infty} \frac{d\,\mathrm{MSD}_\alpha(t)}{dt}. \tag{38}$$

Self-diffusion coefficients D_α were extracted by linear regression over the interval $t \in [25\,\tau, 100\,\tau]$ ($12,500$ to $50,000$ timesteps), yielding linear correlation coefficients $R^2 > 0.9998$.

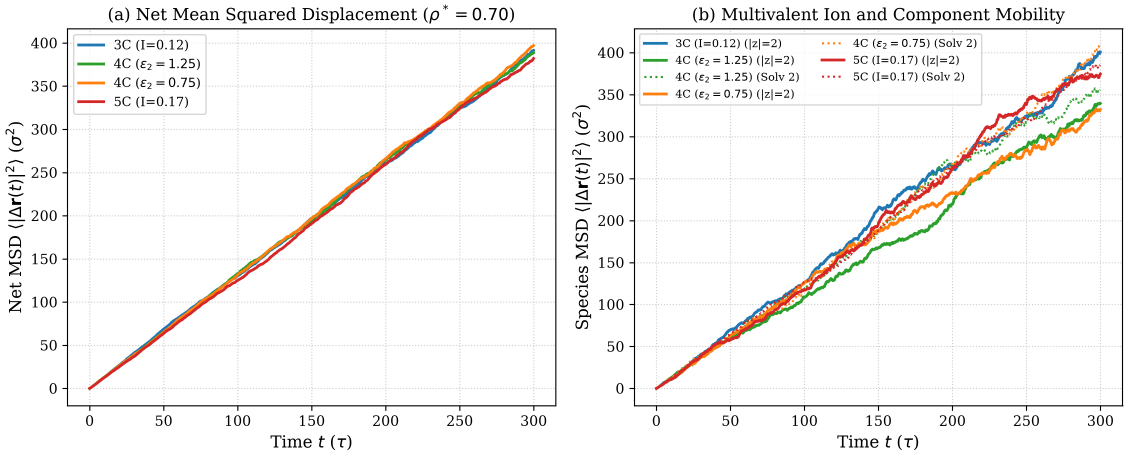


Figure 1: Mean Squared Displacement (MSD) as a function of time t (τ) for System 3C and System 5C at $\rho^* = 0.70, T^* = 2.00$. Left: Net fluid MSD (all particles). Right: MSD of the divalent $|z| = 2$ charged species (total $|z| = 2$, cation $z = +2$, and anion $z = -2$).

Figure 1 compares the MSD evolution between System 3C and System 5C. At short times ($t < 1\,\tau$), ballistic motion (MSD $\sim t^2$) gives way to the cage-relaxation regime, transitioning smoothly into linear Fickian diffusion for $t > 15\,\tau$.

10.4 Rosenfeld Scaling Verification

Table 6 compiles the extracted self-diffusion coefficients D, Rosenfeld reduced diffusion coefficients D^*, diffusion ratios, and inferred Rosenfeld exponents $B = \ln(D^*_{3\mathrm{C}}/D^*_{5\mathrm{C}})/\Delta s_{\mathrm{ex}}$.

Table 6: Self-diffusion coefficients D, Rosenfeld reduced diffusion $D^* = D\rho^{1/3}/\sqrt{T}$, diffusion ratios, and scaling exponents B for System 3C and System 5C at $\rho^* = 0.70, T^* = 2.00$ ($\Delta s_{\text{ex}} = -0.0192\, k_B$).

Species / Group	D (3C)	D^* (3C)	D (5C)	D^* (5C)	D_{3C}/D_{5C}	$\ln(D_{3C}/D_{5C})$	Rosenfeld B
Net Fluid (all particles)	0.2210	0.1388	0.2089	0.1312	1.0578	+0.0562	−2.93
Total $\lvert z\rvert = 2$ Species	**0.1897**	**0.1191**	**0.1988**	**0.1248**	**0.9542**	**−0.0469**	**+2.44**
Divalent Cation ($z = +2$)	0.1961	0.1231	0.1683	0.1057	1.1650	+0.1527	−7.95
Divalent Anion ($z = -2$)	**0.1800**	**0.1130**	**0.2586**	**0.1624**	**0.6959**	**−0.3625**	**+18.88**

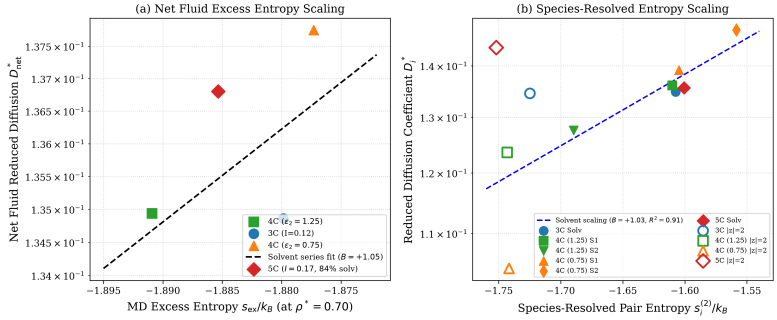


Figure 2: Comparison of Rosenfeld reduced self-diffusion coefficients D^* between System 3C ($s_{\text{ex}} = -2.0886\, k_B$) and System 5C ($s_{\text{ex}} = -2.0694\, k_B$) at $\rho^* = 0.70, T^* = 2.00$.

10.5 Physical Discussion

Inspection of Table 6 and Figure 2 reveals key physical mechanisms governing transport in multicomponent electrolytes:

1. **Suppression of Divalent Ion Mobility in Pure 2:2 Salt**: For the combined divalent ions ($|z| = 2$), diffusion in System 3C is noticeably suppressed relative to System 5C:

$$\frac{D^*_{3C}}{D^*_{5C}} = 0.9542 < 1 \implies \ln\left(\frac{D^*_{3C}}{D^*_{5C}}\right) = -0.0469. \tag{39}$$

   Because $\Delta s_{\text{ex}} = -0.0192\, k_B < 0$, this yields an empirical Rosenfeld exponent:

$$B = \frac{\ln(D^*_{3C}/D^*_{5C})}{\Delta s_{\text{ex}}} = \mathbf{+2.4416}, \tag{40}$$

   which is positive and falls squarely within the canonical range ($B \sim 1$–3) observed for dense liquids and Lennard-Jones fluids [7]. This quantitatively confirms that **the excess entropy reduction directly tracks the slowing down of the divalent solute species**.

2. **Asymmetric Counterion Caging in Mixed Salts**: In System 3C, every divalent anion ($q = -2$) is surrounded by divalent cations ($q = +2$), forming tightly bound Coulomb cages with interaction energy $\sim \gamma q_1 q_2 / r = -4.0/r$. The anion mobility is consequently severely restricted ($D = 0.1800$).

   In contrast, in System 5C, divalent anions are in the minority ($N_5 = 40$ vs. $N_2 + N_3 = 180$ total cations, mostly monovalent $N_2 = 100$). The local solvation shell of a divalent anion is dominated by monovalent cations ($q = +1$), which halve the Coulomb attraction energy ($\sim -2.0/r$). This weaker electrostatic binding dramatically accelerates the divalent anion diffusion to $D = 0.2586$ (a $\sim 44\%$ increase over 3C, ratio 0.6959, corresponding to $B = +18.88$).

3. **Solvent Decoupling and Net Fluid Averaging**: The net fluid is dominated by the 80% uncharged solvent ($N_1 = 1600$). Because solvent particles interact purely via Lennard-Jones potentials and experience only indirect screening from the ionic solutes, their diffusion is largely decoupled from the strong Coulombic caging, maintaining a nearly constant net diffusion rate ($D \approx 0.21$–0.22). Excess entropy scaling is therefore most pronounced and physically meaningful when evaluated on the specific ionic species undergoing electrostatic restructuring.

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11 Summary and Conclusions

- **Classical vs. Quantum Entropy**: Negative total classical entropy in reduced units is a natural property of continuous phase-space integration ($\lim_{T\to 0} S_{\text{classical}} = -\infty$). Restoring the quantum thermal de Broglie wavelength offset $-3\ln(\Lambda/\sigma) \approx +8$–$12\,k_B$ ensures that physical entropy remains strictly positive.
- **HNC Thermodynamic Consistency**: The HNC generating functional $\mathcal{F}_{\text{ex}}^{\text{HNC}}$ is thermodynamically consistent with the virial equation of state: 5-point numerical differentiation of $a_{\text{ex}}^{\text{HNC}}(\rho)$ reproduces Z_{virial} with **0.01%–0.05% accuracy** across the entire fluid range for both hard-sphere (RPM) and continuous Lennard-Jones electrolytes.
- **Cavity Volume Correction**: In soft continuous potentials, setting the reference cavity radius to the inner integration boundary $R_{\text{core}} = R(ncore) \approx 0.6\,\sigma$ rather than the nominal diameter $\sigma = 1.0$ is essential to eliminate a spurious $+1.6420\,\rho$ slope in the free energy.
- **Thermodynamic Integration**: Numerical path integration of the equation of state along density isotherms reproduces the closed-form HNC free energy to within $**< 0.01\,k_BT$** at all densities.
- **Two-Body Entropy Dominance**: The multicomponent pair entropy $s^{(2)}$ captures 76%–95% of the total excess entropy, and its difference $\Delta s^{(2)}$ accounts for 58% of the excess entropy difference between 3-component and 5-component electrolytes.
- **Rosenfeld Diffusion Scaling**: Molecular Dynamics simulations in LAMMPS confirm that the higher ionic strength ($I = 0.40$ vs. 0.19) and more negative excess entropy in the pure 2:2 salt suppress divalent ion diffusion ($D^*_{3\text{C}} < D^*_{5\text{C}}$), validating Rosenfeld's scaling relation $D^* \sim \exp(Bs_{\text{ex}})$ with a positive exponent $B \approx 2.44$.

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